

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Solution: Standard laptop order automation in ServiceNow using Flow Designer

Architecture summary: Requester (browser) → ServiceNow Service Catalog → Request / Requested Item → Approval → Flow Designer flow "Standard laptop task" → Create Catalog Task → Catalog Task assigned to Hardware group.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Requester places the order and approvers/hardware team work in the ServiceNow Service Catalog and platform forms	ServiceNow Service Catalog / Platform UI (browser)
2.	Application Logic-1	Flow "Standard laptop task" (Application: Global, Run as: System User) that automates the process	ServiceNow Flow Designer
3.	Application Logic-2	Service Catalog trigger and Create Catalog Task action that generate the configuration task from the Requested Item	Flow Designer trigger and action
4.	Application Logic-3	Approval of the request through the Approvers section before the task is processed	ServiceNow Approvals
5.	Database	Request (sc_request), Requested Item (sc_req_item), Catalog Task (sc_task), Catalog Item (sc_cat_item) and Group (sys_user_group) tables	ServiceNow platform database
6.	Cloud Database	Data stored in the ServiceNow instance database hosted in the ServiceNow cloud	ServiceNow Cloud instance
7.	File Storage	No separate file storage required; any attachments use the platform's standard attachment storage	ServiceNow attachments (platform)
8.	External API-1	Not used in this project	Not applicable
9.	External API-2	Not used in this project	Not applicable
10.	Machine Learning Model	Not used in this project	Not applicable

11.	Infrastructure (Server / Cloud)	Solution is configured and runs on a ServiceNow cloud instance accessed through a web browser; no local server is required	ServiceNow Cloud (instance)
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is used; the solution is built with the ServiceNow low-code platform	ServiceNow Flow Designer (low-code)
2.	Security Implementations	User authentication, role-based access control (ACLs), approval before processing, and flow executed as System User	ServiceNow roles / ACLs, Approvals
3.	Scalable Architecture	Reusable flow that can be attached to other catalog items and assignment groups; runs on the cloud platform	Flow Designer, ServiceNow Cloud
4.	Availability	The solution is available whenever the ServiceNow cloud instance is available; no separate server to maintain	ServiceNow Cloud
5.	Performance	Tasks are created automatically by the flow after approval, removing manual steps and delays	Flow Designer automation